

Your grade: **100%**

Your latest: **100%** • Your highest: **100%** • To pass you need at least 75%. We keep your highest score.

Next item →

1. Which of these statements are true? Select any/all that apply:

1 / 1 point

☒ Every set of Euler angles corresponds to a unit quaternion.

✔ Nice work

Correct! Any set of Euler angles represents a rotation which can also be represented by a unit quaternion.

☒ Every unit quaternion has an associated 3x3 rotation matrix.

✔ Nice work

Correct! Every quaternion corresponds to a 3x3 rotation matrix.

☐ Every 3x3 matrix represents a 3D rotation.

2. Which of these are valid rotation matrices? Select any/all that apply:

2 / 2 points

☐ $C_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$

☒ $C_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$

✔ Nice work

Correct! $C_2 C_2^T = \mathbf{I}$ holds and $\det C_2 = 1$.

☐ $C_4 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

☒ $C_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

3. Localization can be performed on board a vehicle by integrating the rotational velocities and linear accelerations measured by an IMU. Assuming that the IMU measurement noise is drawn from a normal distribution, what will the pose estimation error look like?

1 / 1 point

- ☒ The vehicle pose estimation error will grow with time.
- ☐ The vehicle pose estimate error will remain within a bounded interval.
- ☐ The vehicle pose estimate error will decrease with time.

✓ Nice work

Correct! Since we are integrating noisy measurements, the error will build up over time.

4. Each GPS satellite transmits a signal that encodes:

1 / 1 point

- ☐ The receiver's position and time of signal transmission
- ☒ The satellite's position and time of signal transmission.

✓ Nice work

Correct! This information can be used to calculate the vehicle's position.

5. Which of these systems provides the most accurate positioning measurement?

1 / 1 point

- ☒ RTK GPS
- ☐ GPS
- ☐ DGPS

✔ Nice work

Correct! RTK uses phase of the GPS carrier signal to provide centimetre-level accuracy.

6. What is the minimum number of GPS satellites required to estimate the 3D position of a vehicle through trilateration?

2 / 2 points

4

✔ Nice work

Correct! A minimum of four satellites is required to unambiguously calculate the vehicle's 3D position.